

CI/CD WORKFLOW FOR QA

1. Introduction to CI/CD in QA

Continuous Integration and Continuous Deployment (CI/CD) is a modern software development practice where code changes are automatically built, tested, and deployed. In QA, CI/CD ensures that automated tests are executed continuously to maintain software quality.

2. Tools Used

- Version Control: GitHub
- CI/CD Tools: GitHub Actions
- Automation Tool: Selenium WebDriver
- Reporting Tools: Extent Reports

3. Workflow Overview

The CI/CD pipeline integrates Selenium automation scripts into the development lifecycle. The process ensures that every code change is tested automatically.

4. Pipeline Proces

Step 1: Code Commit (Trigger Point)

- A developer pushes code to the GitHub repository
- This action acts as the **trigger** for the CI/CD pipeline

Step 2: Pipeline Triggered

- CI/CD tool (Jenkins or GitHub Actions) detects the change
- Pipeline starts automatically

Step 3: Environment Setup

- System installs required dependencies:

- Browser (Chrome/Firefox)
- WebDriver (ChromeDriver)
- Programming language (Java/Python)
- Test environment is prepared

Step 4: Build Phase

- Application build is created (if required)
- Latest code is fetched from repository

Step 5: Test Execution

- Selenium scripts are executed automatically
- Test cases run on the application

Example:

- Login test
- Dashboard test
- Navigation test

5. Test Reports Generation

After execution, reports are automatically generated.

Types of Reports:

- Pass/Fail status
- Execution time
- Screenshots (for failed tests)
- Logs and error messages

Tools Used:

- Allure Report
- Extent Report

Report Flow:

1. Test execution completes

2. Results collected
3. Report generated automatically
4. Report saved in pipeline artifacts

6. Result Handling

If Tests Pass:

- Pipeline continues
- Application is deployed to staging/production

If Tests Fail:

- Pipeline stops
- Bugs are reported
- Developer is notified

7. Notification System

- Email notifications sent to team
- GitHub/Jenkins dashboard updated
- QA team reviews failed test cases

8. Benefits of CI/CD in QA

- Faster testing process
- Early bug detection
- Continuous feedback
- Improved software quality
- Reduced manual effort

9. Conclusion

Integrating Selenium automation with CI/CD pipelines ensures efficient and reliable testing. It allows teams to detect issues early and maintain high-quality software in a fast-paced development environment.

